

- Wait
- Wait
- Wait
- Take at bedtime to help reduce daytime sedation
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 10–20 mg/day (oral or intramuscular)
- 6–12 mg olanzapine/25–50 mg fluoxetine (olanzapine/fluoxetine combination)
- 150–300 mg/2 weeks or 300–405 mg/4 weeks (see Olanzapine Pamoate after Pearls for dosing and use)

Dosage Forms

- Tablet 2.5 mg, 5 mg, 7.5 mg, 10 mg, 15 mg, 20 mg
- Orally disintegrating tablet 5 mg, 10 mg, 15 mg, 20 mg
- Intramuscular formulation 5 mg/mL, each vial contains 10 mg (available in some countries)
- Depot 210 mg, 300 mg, 405 mg
- Olanzapine/fluoxetine combination capsule (mg equivalent olanzapine/mg equivalent fluoxetine) 3 mg/25 mg, 6 mg/25 mg, 6 mg/50 mg, 12 mg/25 mg, 12 mg/50 mg

How to Dose

- Initial 5–10 mg once daily orally; increase by 5 mg/day once a week until desired efficacy is reached; maximum approved dose is 20 mg/day

- For intramuscular formulation, recommended initial dose 10 mg; second injection of 5–10 mg may be administered 2 hours after first injection; maximum daily dose of olanzapine is 20 mg, with no more than 3 injections per 24 hours
- For olanzapine/fluoxetine combination, recommended initial dose 6 mg/25 mg once daily in evening; increase dose based on efficacy and tolerability; maximum generally 18 mg/75 mg



Dosing Tips –Oral

- * **More may be more:** raising usual dose above 15 mg/day can be useful for acutely ill and agitated patients and some treatment-resistant patients, gaining efficacy without many more side effects
- * Some heroic uses for patients who do not respond to other antipsychotics can occasionally justify dosing over 30 mg/day and short-term up to 90 mg/day
- For high doses in treatment-resistant or violent patients, monitor therapeutic drug levels and target generally higher than the usual range of 5–75 mg/mL (i.e., greater than 125 mg/mL), but keep below the toxic range associated with QTc prolongation (700–800 mg/mL)
- See also the Switching section, after Pearls, for initiating both oral and long-acting injectable
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term

(one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly

- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Clearance of olanzapine is slightly reduced in women compared to men, so women may need lower doses than men
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- * Olanzapine intramuscularly can be given short-term, both to initiate dosing with oral olanzapine or another oral antipsychotic and to treat breakthrough agitation in patients maintained on oral antipsychotics
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$
- For treatment resistance, consider obtaining plasma drug levels to guide dosing above recommended levels

Overdose

- Rarely lethal in monotherapy overdose; sedation, slurred speech

Long-Term Use

- Approved to maintain response in long-term treatment of schizophrenia
- Approved for long-term maintenance in bipolar disorder

- Often used for long-term maintenance in various behavioral disorders
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- See Switching section of individual agents for how to stop olanzapine
- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Metabolites are inactive
- Parent drug has 21–54 hour half-life
- Substrate for CYP450 1A2 and 2D6
- Food does not affect absorption



Drug Interactions

- May increase effect of antihypertensive agents
- May antagonize levodopa, dopamine agonists

- Dose may need to be lowered if given with CYP450 1A2 inhibitors (e.g., fluvoxamine); raised if given in conjunction with CYP450 1A2 inducers (e.g., cigarette smoke, carbamazepine)

Other Warnings/Precautions

- Olanzapine is associated with a rare but serious skin condition known as drug reaction with eosinophilia and systemic symptoms (DRESS). DRESS may begin as a rash but can progress to other parts of the body and can include symptoms such as fever, swollen lymph nodes, swollen face, inflammation of organs, and an increase in white blood cells known as eosinophilia. In some cases, DRESS can lead to death. Clinicians prescribing olanzapine should inform patients about the risk of DRESS; patients who develop a fever with rash and swollen lymph nodes or swollen face should seek medical care. Patients are not advised to stop their medication without consulting their prescribing clinician
- Monitor patients for activation of suicidal ideation, especially children and adolescents
- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Use with caution in patients with prostatic hypertrophy, angle-closure glaucoma, paralytic ileus
- Patients receiving the intramuscular formulation of olanzapine should be observed closely for hypotension

- Intramuscular formulation is not generally recommended to be administered with parenteral benzodiazepines; if patient requires a parenteral benzodiazepine it should be given at least 1 hour after intramuscular olanzapine
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and olanzapine should be used cautiously in patients at risk for aspiration pneumonia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If there is a known risk of angle-closure glaucoma (intramuscular formulation)

- If patient has unstable medical condition (e.g., acute myocardial infarction, unstable angina pectoris, severe hypotension and/or bradycardia, sick sinus syndrome, recent heart surgery) (intramuscular formulation)
- If there is a proven allergy to olanzapine

Special Populations

Renal Impairment

- No dose adjustment required for oral formulation
- Not removed by hemodialysis
- For intramuscular formulation, consider lower starting dose (5 mg)

Hepatic Impairment

- May need to lower dose
- Patients with liver disease should have liver function tests a few times a year
- For moderate to severe hepatic impairment, starting oral dose 5 mg; increase with caution
- For intramuscular formulation, consider lower starting dose (5 mg)

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Some patients may tolerate lower doses better
- Increased incidence of stroke
- For intramuscular formulation, recommended starting dose is 2.5–5 mg; a second injection of 2.5–5 mg may be administered 2 hours after first injection; no more than 3 injections should be administered within 24 hours
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events
- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic



Children and Adolescents

- Approved for use in schizophrenia and manic/mixed episodes (ages 13 and older for both)
- Clinical experience and early data suggest olanzapine is probably safe and effective for behavioral disturbances in children and adolescents
- Children and adolescents using olanzapine may need to be monitored more often than adults and may tolerate lower doses better
- Intramuscular formulation has not been studied in patients under 18 and is not recommended for use in this population

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding

- Psychotic symptoms may worsen during pregnancy, and some form of treatment may be necessary
- Early findings of infants exposed to olanzapine in utero currently do not show adverse consequences
- When administered to pregnant rats and rabbits, olanzapine was not teratogenic at doses 9 and 30 times the maximum recommended human dose; some fetal toxicities were observed at these doses
- Olanzapine may be preferable to anticonvulsant mood stabilizers if treatment is required during pregnancy
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypicalantipsychotic>

Breast Feeding

- Unknown if olanzapine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed while on olanzapine should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- * Some cases of psychosis and bipolar disorder refractory to treatment with other antipsychotics
- * Often a preferred augmenting agent in bipolar depression or treatment-resistant unipolar depression
- * Patients needing rapid onset of antipsychotic action without drug titration
- Patients switching from intramuscular olanzapine to an oral preparation

Potential Disadvantages

- Patients concerned about gaining weight
- * Patients with diabetes mellitus, obesity, and/or dyslipidemia

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depressed mood and mania)
- Aggressive symptoms

Pearls

- Head-to-head study in schizophrenia suggests greater effectiveness (i.e., lower dropouts of all causes) at moderately high doses

compared to some other atypical and conventional antipsychotics at moderate doses

- Same head-to-head study in schizophrenia suggests greater efficacy but greater metabolic side effects compared to some other atypical and conventional antipsychotics
- Well accepted for use in schizophrenia and bipolar disorder, including difficult cases
- * Documented utility in treatment-refractory cases, especially at higher doses
- * Documented efficacy as augmenting agent to SSRIs (fluoxetine) in nonpsychotic treatment-resistant major depressive disorder
- * Documented efficacy in bipolar depression, especially in combination with fluoxetine
- Combination of olanzapine with the opioid antagonist samidorphan in late-stage clinical development to mitigate weight gain
- For treatment-refractory cases, consider dosing guided by plasma drug levels and tolerability, with minimum therapeutic level of 40 ng/mL and point of futility of 200 ng/mL; many who don't respond to lower levels have a clinical response at plasma drug levels above 125 ng/mL (may require oral dosing at or above 40 mg/day to attain these plasma drug levels)
- More weight gain than many other antipsychotics –does not mean every patient gains weight
- Motor side effects unusual at low to mid-doses

- Less sedation than for some other antipsychotics, more than for others
- * Controversial as to whether olanzapine has more risk of diabetes and dyslipidemia than other antipsychotics
- Cigarette smoke can decrease olanzapine levels and patients may require a dose increase if they begin or increase smoking
- * A short-acting intramuscular dosage formulation is available
- Long-acting intramuscular dosage formulation is also approved
- Patients with inadequate responses to atypical antipsychotics may benefit from determination of plasma drug levels and, if low, a dosage increase even beyond the usual prescribing limits
- Patients with inadequate responses to atypical antipsychotics may also benefit from a trial of augmentation with a conventional antipsychotic or switching to a conventional antipsychotic
- However, long-term polypharmacy with a combination of a conventional antipsychotic with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring

- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

Olanzapine Pamoate

Vehicle	Water
Tmax	3–4 days
T1/2 with multiple dosing	30 days
Time to reach steady state	3 months
Able to be loaded	Yes
Dosing schedule (maintenance)	2 weeks or 4 weeks
Injection site	Intramuscular gluteal
Needle gauge	19
Dosage forms	210 mg, 300 mg, 405 mg
Injection volume	150 mg/mL (range 1.0–2.7 mL)

Usual Dosage Range

- 150–300 mg/2 weeks or 300–405 mg/4 weeks

How to Dose

- Conversion from oral: dose should be loaded during the initial 8 weeks based on the prior stable oral dose of olanzapine

Daily oral olanzapine dose	LAI dose: first 8 weeks	LAI dose: after 8 weeks
10 mg	210 mg/2 weeks OR 405 mg/4 weeks	150 mg/2 weeks OR 300 mg/4 weeks
15 mg	300 mg/2 weeks	210 mg/2 weeks OR 405 mg/4 weeks
20 mg	300 mg/2 weeks	300 mg/2 weeks

- Oral supplementation may be needed if adequate loading is not used
- Maximum dose 300 mg/2 weeks

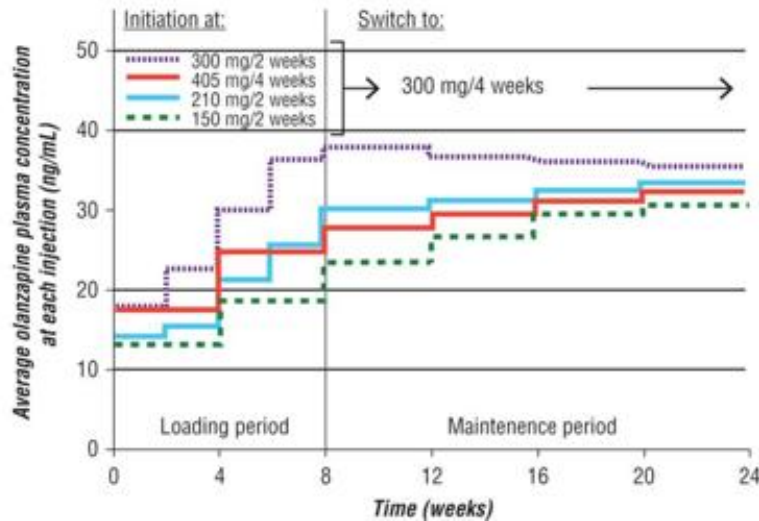
Dosing Tips

- With LAIs, the absorption rate constant is slower than the elimination rate constant, thus resulting in “flip-flop” kinetic –i.e., time to steady state is a function of absorption rate, while concentration at steady state is a function of elimination rate
- The rate-limiting step for plasma drug levels for LAIs is not drug metabolism, but rather slow absorption from the injection site

- In general, 5 half-lives of any medication are needed to achieve 97% of steady-state levels
- The long half-lives of depot antipsychotics mean that one must either adequately load the dose (if possible) or provide oral supplementation
- The failure to adequately load the dose leads either to prolonged cross-titration from oral antipsychotic or to sub-therapeutic antipsychotic plasma levels for weeks or months in patients who are not receiving (or adhering to) oral supplementation
- Because plasma antipsychotic levels increase gradually over time, dose requirements may ultimately decrease from initial; obtaining periodic plasma levels can be beneficial to prevent unnecessary plasma level creep
- The time to get a blood level for patients receiving LAI is the morning of the day they will receive their next injection
- Advantages: efficacy advantage of oral olanzapine
- Disadvantages: 3-hour post-injection monitoring required due to risk (0.2%) of post-injection delirium from vascular breach
- Response threshold is generally 21 ng/mL; plasma levels greater than 176 ng/mL are generally not well tolerated

Switching from Oral Antipsychotics to Olanzapine Pamoate

Expected Olanzapine Levels Without Oral Coverage



- Discontinuation of oral antipsychotic can begin immediately if adequate loading is pursued
- How to discontinue oral formulations
 - Down-titration is not required for: amisulpride, aripiprazole, brexpiprazole, cariprazine, olanzapine, paliperidone ER
 - 1-week down-titration is required for: iloperidone, lurasidone, risperidone, ziprasidone, asenapine, quetiapine
 - 4+-week down-titration may be required for: clozapine

The Art of Switching

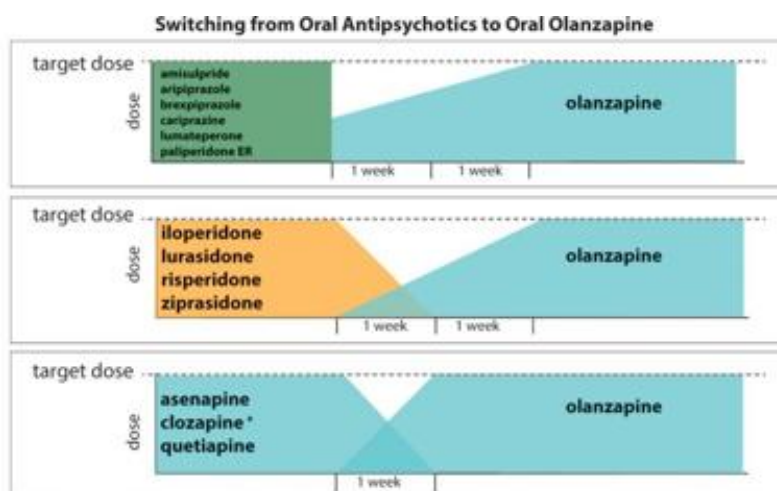


Switching from Oral Antipsychotics to Oral Olanzapine

- With amisulpride, aripiprazole, brexpiprazole, cariprazine, lumateperone, and paliperidone ER, immediate stop is possible; begin olanzapine at middle dose

- With risperidone, ziprasidone, iloperidone, and lurasidone, begin olanzapine gradually, titrating over at least 2 weeks to allow patients to become tolerant to the sedating effect

*May need to taper clozapine slowly over 4 weeks or longer



Suggested Reading

Citrome L. Adjunctive aripiprazole, olanzapine, or quetiapine for major depressive disorder: an analysis of number needed to treat, number needed to harm, and likelihood to be helped or harmed . *Postgrad Med* 2010 ;**122** (4):39 –48 .

Detke HC , Zhao F , Garhyan P , Carlson J , McDonnell D . Dose correspondence between olanzapine long-acting injection and oral olanzapine: recommendations for switching . *Int Clin Psychopharmacol* 2011 ;**26** :35 –42 .

Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of

adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Nasrallah HA . Atypical antipsychotic-induced metabolic side effects: insights from receptor-binding profiles . *Mol Psychiatry* 2008 ;**13** (1):27 – 35 .

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Smith LA , Cornelius V , Warnock A , Tacchi MJ , Taylor D . Pharmacological interventions for acute bipolar mania: a systematic review of randomized placebo-controlled trials . *Bipolar Disord* 2007 ;**9** (6):551–60 .

Oxazepam

Therapeutics

Brands

- Serax
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (anxiolytic)

Commonly Prescribed for

(bold for FDA approved)

- **Anxiety**
- **Anxiety associated with depression**
- **Alcohol withdrawal**
- Catatonia

How the Drug Works

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibits neuronal activity presumably in amygdala-centered fear circuits to provide therapeutic benefits in anxiety disorders

How Long Until It Works

- Some immediate relief with first dosing is common; can take several weeks with daily dosing for maximal therapeutic benefit

If It Works

- For short-term symptoms of anxiety –after a few weeks, discontinue use or use on an “as-needed” basis
- For chronic anxiety disorders, the goal of treatment is complete remission of symptoms as well as prevention of future relapses
- For chronic anxiety disorders, treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- For long-term symptoms of anxiety, consider switching to an SSRI or SNRI for long-term maintenance

- If long-term maintenance with a benzodiazepine is necessary, continue treatment for 6 months after symptoms resolve, and then taper dose slowly
- If symptoms reemerge, consider treatment with an SSRI or SNRI, or consider restarting the benzodiazepine; sometimes benzodiazepines have to be used in combination with SSRIs or SNRIs for best results

If It Doesn't Work

- Consider switching to another agent or adding an appropriate augmenting agent
- Consider psychotherapy, especially cognitive behavioral psychotherapy
- Consider presence of concomitant substance abuse
- Consider presence of oxazepam abuse
- Consider another diagnosis, such as a comorbid medical condition

Best Augmenting Combos for Partial Response or Treatment Resistance

- Benzodiazepines are frequently used as augmenting agents for antipsychotics and mood stabilizers in the treatment of psychotic and bipolar disorders
- Benzodiazepines are frequently used as augmenting agents for SSRIs and SNRIs in the treatment of anxiety disorders
- Not generally rational to combine with other benzodiazepines

- Caution if using as an anxiolytic concomitantly with other sedative hypnotics for sleep

Tests

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent

Side Effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors
- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal
- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

- * Sedation, fatigue, depression
- * Dizziness, ataxia, slurred speech, weakness
- * Forgetfulness, confusion
- * Hyperexcitability, nervousness

- Rare hallucinations, mania
- Rare hypotension
- Hypersalivation, dry mouth



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with CNS depressants in overdose
- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Especially at initiation of treatment or when dose increases
- Tolerance often develops over time

What to Do About Side Effects

- Wait
- Wait
- Wait